

Course Name:	Artificial Intelligence Laboratory	Semester:	VI
Date of Performance:	9/04/ 2026	DIV/ Batch No:	C3
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Experiment No: 8

Title: Implementation of Bayesian Network Using Python libraries.

Aim: Understand the inferences in uncertain scenario

Objectives of the Experiment:

By the end of this experiment, students will be able to:

1. Understand which python library is to be implemented for Bayesian Problems
2. How to use them to solve various problems.

COs to be achieved:

CO3: Represent and formulate the knowledge to solve the problems using various reasoning techniques

Books/ Journals/ Websites referred:

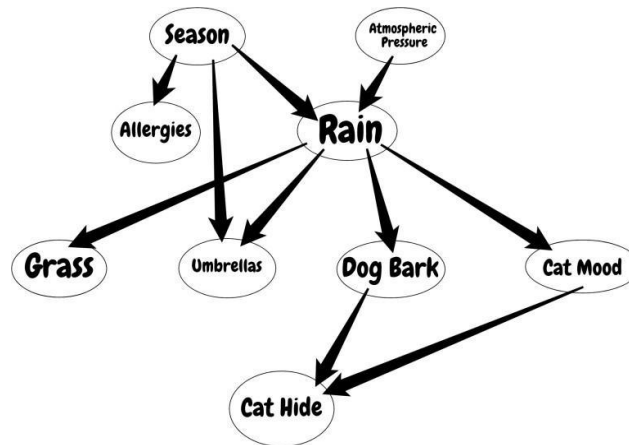
1. “Artificial Intelligence: a Modern Approach” by Russel and Norving, Pearson education Publications
2. “Artificial Intelligence” By Rich and knight, Tata Mcgraw Hill Publications
3. <https://machinelearningmastery.com/introduction-to-bayesian-belief-networks/>, last retried on April 02,2025
4. <https://towardsdatascience.com/introduction-to-bayesian-belief-networks-c012e3f59f1b> , last retried on April 02,2025

Theory: Bayesian networks:

A Bayesian network (also known as a Bayes network, belief network, or decision network) is a probabilistic graphical model that represents a set of variables and their conditional dependencies via a directed acyclic graph (DAG). Bayesian networks are ideal for taking an event that occurred and predicting the likelihood that any one of several possible known causes was the contributing factor. For example, a Bayesian network could represent the probabilistic relationships between diseases and symptoms. Given symptoms, the network can be used to compute the probabilities of the presence of various diseases.

Efficient algorithms can perform inference and learning in Bayesian networks. Bayesian networks that model sequences of variables (e.g. speech signals or protein sequences) are called dynamic Bayesian networks. Generalizations of Bayesian networks that can represent and solve decision problems under uncertainty are called influence diagrams.

Bayesian Belief Network or Bayesian Network or Belief Network is a Probabilistic Graphical Model (PGM) that represents conditional dependencies between random variables through a Directed Acyclic Graph (DAG).



Bayesian Networks are applied in many fields. For example, disease diagnosis, optimized web search, spams filtering, gene regulatory networks, etc. And this list can be extended. The main objective of these networks is trying to understand the structure of causality relations. To clarify this, let's consider a disease diagnosis problem. With given symptoms and their resulting disease, we construct our Belief Network and when a new patient comes, we can infer which disease or diseases may have the new patient by providing probabilities for each disease. Similarly, these causality relations can be constructed for other problems and inference

techniques can be applied to interesting results.

As you would understand from the formula, to be able to calculate the joint distribution we need to have conditional probabilities indicated by the network. But further that if we have the joint distribution, then we can start to ask interesting questions. For example, in the first example, we ask for the probability of “RAIN” if “SEASON” is “WINTER” and “DOG BARK” is “TRUE”.

Activity

$$P(X|Y) = \frac{P(X, Y)}{P(Y)}$$

Chosen Problem: For the problem, define the network, calculate the probabilities and query the system.

Problem Statement: Implementation of Bayesian Network for Student Performance Prediction. Student academic performance depends on multiple uncertain factors such as study habits, attendance, sleep, and stress levels. These factors interact in complex ways, making it difficult to predict outcomes deterministically. This experiment models the probabilistic relationships between these factors using a Bayesian Belief Network (BBN) to compute the likelihood of a student performing well or poorly based on observed evidence.

Link to your colab notebook if you have implemented your own problem:

[16010123325-ShreyansTatiya-AI](https://colab.research.google.com/drive/16010123325-ShreyansTatiya-AI)

Query 1:

Computations for query 1:

Query 1: Prior probability of Performance (Pass/Fail)

[6]
✓ 0s

```
query1_result = infer.query(variables=['Performance'])
print(query1_result)
```

```

+-----+-----+
| Performance | phi(Performance) |
+=====+=====+
| Performance(Pass) | 0.4790 |
+-----+-----+
| Performance(Fail) | 0.5210 |
+-----+-----+
  
```

Query 2:

Computations for query 2:

Query 2: Probability of Performance (Pass/Fail) given StudyHours = High

[7]
✓ 0s

```
query2_result = infer.query(variables=['Performance'], evidence={'StudyHours': 'High'})
print(query2_result)
```

```

+-----+-----+
| Performance | phi(Performance) |
+=====+=====+
| Performance(Pass) | 0.7100 |
+-----+-----+
| Performance(Fail) | 0.2900 |
+-----+-----+
  
```

Query 3:

Computations for query 3:

Query 3: Probability of Performance (Pass/Fail) given Attendance = Good and Stress = Low

```
[8]
✓ Os query3_result = infer.query(variables=['Performance'], evidence={'Attendance': 'Good', 'Stress': 'Low'})
      print(query3_result)
```

Performance	phi(Performance)
Performance(Pass)	0.6600
Performance(Fail)	0.3400

Query 4:
Computations for query 4:

Query 4: Probability of Stress (High/Low) given Performance = Fail

```
[9]
✓ Os query4_result = infer.query(variables=['Stress'], evidence={'Performance': 'Fail'})
      print(query4_result)
```

Stress	phi(Stress)
Stress(High)	0.5873
Stress(Low)	0.4127

Query 5:
Computations for query 5:

Query 5: Probability of Performance (Fail) given StudyHours = Low, Attendance = Poor, Sleep = Less

```
[10]
✓ Os query5_result = infer.query(variables=['Performance'], evidence={'StudyHours': 'Low', 'Attendance': 'Poor', 'Sleep': 'Less'})
      print(query5_result)
```

Performance	phi(Performance)
Performance(Pass)	0.1200
Performance(Fail)	0.8800

The Bayesian Network successfully models the uncertainty in student performance prediction. Each query demonstrates how different factors influence the outcome and how probabilities are updated when new evidence is introduced.

Post Lab Subjective/Objective type Questions:

Question: Answer in brief

1. Explain the significance of Bayesian Networks in AI. How do they help in decision-making under uncertainty?

Ans. Bayesian Networks are important in AI because they provide a structured way to represent and reason with uncertainty. Instead of relying on fixed rules, they model real-world situations where outcomes are influenced by multiple uncertain factors.

They help in decision-making by calculating the probability of different outcomes based on available evidence. As new information is observed, the network updates its beliefs using probability theory. This makes Bayesian Networks highly useful in applications like diagnosis, prediction, and risk analysis where complete information is not always available.

2. What are the main components of a Bayesian Network? Explain each briefly.

Ans. A Bayesian Network is a probabilistic graphical model composed of two main elements: a Directed Acyclic Graph (DAG) representing causal relationships (structure) and Conditional Probability Tables (CPTs) providing numerical data. Together, they represent variables, dependencies, and uncertainties, allowing for inference and prediction

3. What is Variable Elimination?

Ans. Variable Elimination is a method used to perform efficient inference in Bayesian Networks. Instead of computing the entire joint probability distribution, it removes irrelevant variables step by step.

This is done by combining and simplifying probability factors and summing out variables that are not part of the query. As a result, it reduces computational complexity and allows faster calculation of required probabilities.

4. Suppose you have a Bayesian Network with three variables: Disease, Test Result, and Symptoms. Explain how you would use conditional probabilities to determine the likelihood that a patient has the disease given a positive test result.

Ans. To find the probability of a disease given a positive test result, we apply **Bayes' Theorem**, which allows us to reverse conditional probabilities.

We calculate this by taking the probability of getting a positive test when the disease is present and multiplying it by the prior probability of the disease. Then, we divide this by the overall probability of obtaining a positive test result.

Symptoms can also be included as additional evidence to further refine the probability. This approach helps in estimating how likely it is that the patient actually has the disease based on observed test results and known probabilities.

Conclusion:

I learnt about the application of Bayesian Belief Network using python libraries, and how to visualise and effectively use it to solve problems and get meaningful outcomes.